AADHAV INTECH
GST No: 33BORPA6650L1ZO
No: 60,3rd Street, Kasi Estate,
West Jafferkhanpet, Chennai – 600083.
Contact Number: 044 4204 1270
E Mail: service@aadhavintech.com

Utilities & Installation Requirements for Continuous Emission Monitoring Station with PM

1. Operating Conditions

- AC Room require for CEMS panel. (10 L x 10 B x 10 H) feet
- The installation site shall be **free from vibration, excessive dust, heat, and electromagnetic interference**.
- The instrument **must not be exposed to direct sunlight**.
- Adequate environmental & Lighting protection shall be ensured to maintain reliable performance.

---

2. Power Supply Requirements

- **UPS power supply is mandatory**.
- Incoming power supply to the UPS:
  - Voltage: **230 V AC with Neutral to Earth < 1 Volt.**
  - Protection: **6 Amps MCB**
- **Minimum UPS capacity: 3 kVA**
- Instrument operating power:
  - **230 V ±10%, 50/60 Hz**
  - **Load: 6 Amps**
- Near the CEMS panel Unit:
  - **2 Nos. of 6 Amps UPS-backed power spike sockets** shall be provided.

---

3. IoT Datalogger Requirements

- UPS-backed power supply shall be provided for the **IoT Datalogger**.
- A dedicated permanent LAN connection shall be provided without firewall or network security restrictions to ensure continuous data transmission.
- If a LAN connection cannot be provided, a **multi-SIM dongle with active SIM cards (Airtel / Jio / Vodafone)** shall be arranged to ensure uninterrupted internet connectivity for transferring data to the **TNPCB server**.

AADHAV INTECH
GST No: 33BORPA6650L1ZO
No: 60,3rd Street, Kasi Estate,
West Jafferkhanpet, Chennai – 600083.
Contact Number: 044 4204 1270
E Mail: service@aadhavintech.com

4. General Installation Requirements

4.1 Access & Safety

- A suitable working platform with safety railing shall be provided near the stack sampling point.
- The platform should allow minimum two persons to work safely.

4.2 Stack Mechanical & Welding Requirements for SPM

- Sample flanges shall be welded 180° straight-to-straight on:
  - Transmitter side
  - Receiver side
    (Total: 2 Nos. of stack flanges)
    → Refer attached drawings for flange dimensions and fixing details.
- 2 Nos. of 'U' flanges are required at the chimney stack for Transmitter and Receiver installation. *(These will be supplied along with the instrument.)*
- 2 Nos. of Weather Shelters shall be provided for the Transmitter and Receiver units. *(These will be supplied along with the instrument.)*
- Weather Shelter Mounting arrangement - For the installation of the weather shelters, ¾" MS L-angle supports shall be welded on both sides of the transmitter and receiver flanges to provide adequate structural support and secure mounting. Reference photographs are enclosed for clarity and implementation guidance.

4.3 Instrument Purge Air Requirements

- **½" Instrument air pipeline (GI or MS)** with:
  - Pressure: > 5 kg/cm²
  - Isolation valve
  - Purge Air Filter
- **Instrument air** shall be:
  - Regulated
  - Free from dust, moisture, and oil mist
- **Air purging connection** shall be provided near the chimney platform at the top of the PM Transmitter and Receiver flanges on both sides to facilitate proper operation and maintenance. *(Refer to the attached airline filter photos for reference.)*

5. Electrical & Cabling Requirements

- A **weatherproof junction box** shall be installed near the Transmitter or Receiver unit. *(These will be supplied along with the instrument.)*
- **Instrument Signal cable** routing:
  - Approx. 50 meters signal cable from junction box to SPM Display Unit. *(These will be supplied along with the instrument.)*
- **Cable trays** shall be provided:
  - Between Transmitter, Receiver, and Junction Box
  - From Junction Box to Display Unit
- **Output Signal Cable** Specification:
  - 2 Core, Multi-strand
  - 1 Sq.mm
  - Armoured & Shielded
  - Signal cables must not be routed near high-power cables to avoid interference or signal loss.

---

6. Data Communication

- From the CEMS panel Unit, the RS-485 output signal cable shall be connected to the IoT Datalogger.
- Purpose: SPCB & CPCB data logging software connectivity.

---

7. Lightning Protection

- **Spike arrester** shall be installed at the **top of the stack** for lightning protection.

---

8. PM Transmitter and Receiver flange units' installation position at Monitored stack

8.1 Sampling Port at Stack: Sampling ports shall be provided by the user to fit the LM318 to the stack. The sampling ports are fitted to the stack as shown in the following figure. The sampling ports are fitted exactly 180° to each other on the

AADHAV INTECH
GST No: 33BORPA6650L1ZO
No: 60,3rd Street, Kasi Estate,
West Jafferkhanpet, Chennai – 600083.
Contact Number: 044 4204 1270
E Mail: service@aadhavintech.com

stack. To ensure laminar flow, sampling ports shall be located at minimum 8 times chimney diameter downstream and 2 times upstream. This will keep the

flow away from any disturbance. For a rectangular cross section, the equivalent diameter (D_e) shall be calculated by using following equation to determine up stream, downstream distances.

$$D_e = \frac{2LW}{L+W}$$

Where L =Length in m, W= width in m. The stack flanges diagram is also shown in the figure.

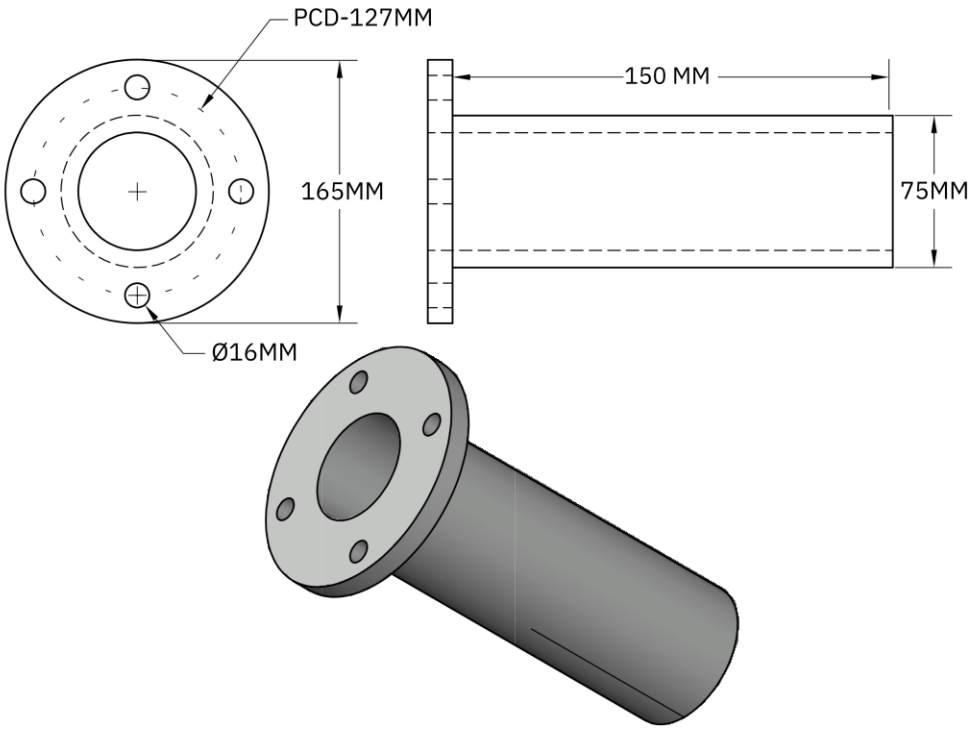


Fig. 1. Flange dimensions for Transmitter & Receiver of Stack side

AADHAV INTECH
GST No: 33BORPA6650L1ZO
No: 60,3rd Street, Kasi Estate,
West Jafferkhanpet, Chennai – 600083.
Contact Number: 044 4204 1270
E Mail: service@aadhavintech.com

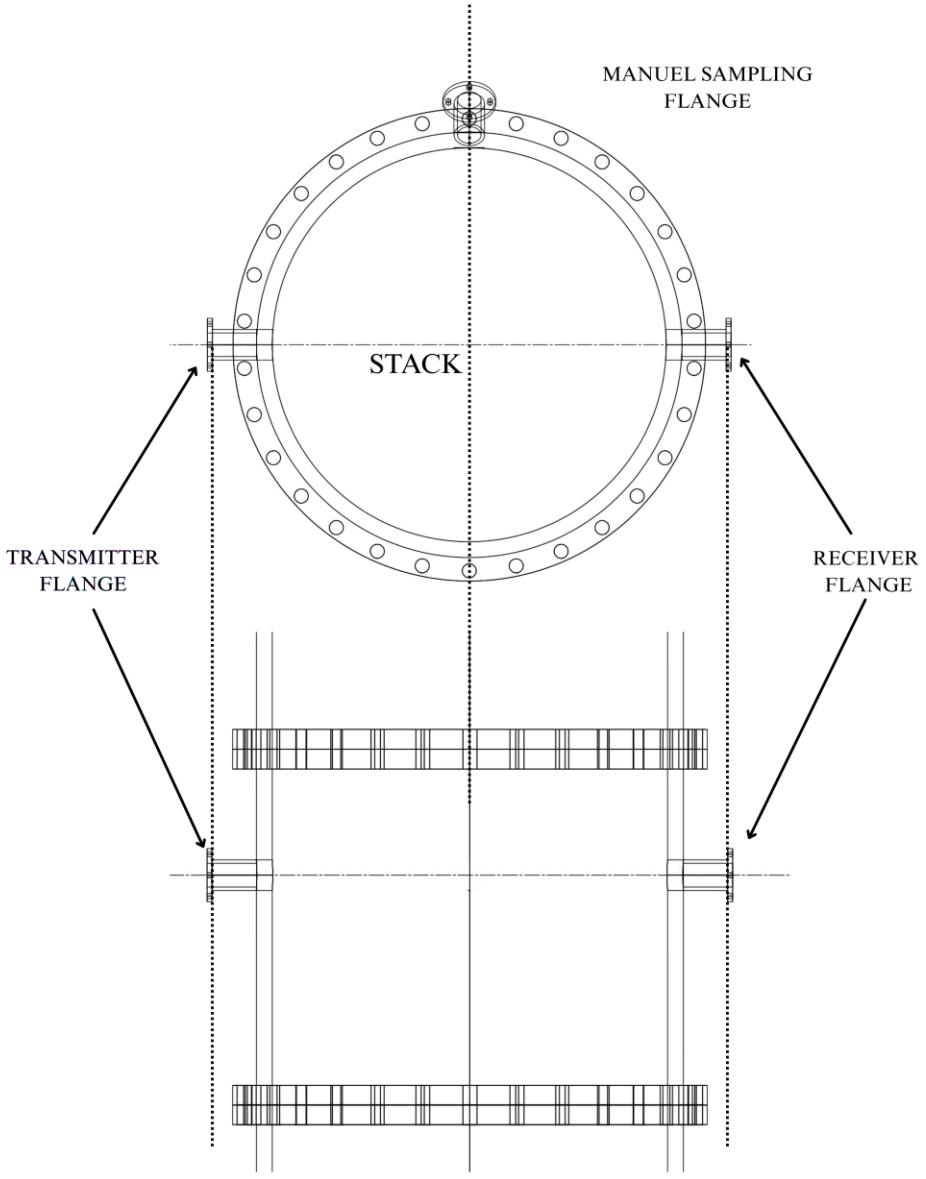


Fig. 2. Stack flanges straight to straight 180^0 position

AADHAV INTECH
GST No: 33BORPA6650L1ZO
No: 60,3rd Street, Kasi Estate,
West Jafferkhanpet, Chennai – 600083.
Contact Number: 044 4204 1270
E Mail: service@aadhavintech.com

9. CEMS Sampling Flange Installation Position at Monitored Stack

9.1 The **CEMS sampling probe** flange shall be welded at a 45° angle at the top of the monitored stack at the specified marked location.
Reference: Fig. 3. An example installation image is attached for reference.

9.2 The **sampling tip filter probe** *(to be supplied along with the instrument)* shall be installed on the welded sampling flange support.

9.3 The **Heated Sampling Hose** (30 meters) *(to be supplied along with the instrument)* shall be laid from the sampling point at the top of the stack to the CEMS panel installation location (AC room) at the bottom of the stack.

9.4 **100 mm supporting cable** tray shall be installed to provide proper routing and mechanical support for the heated sampling hose.

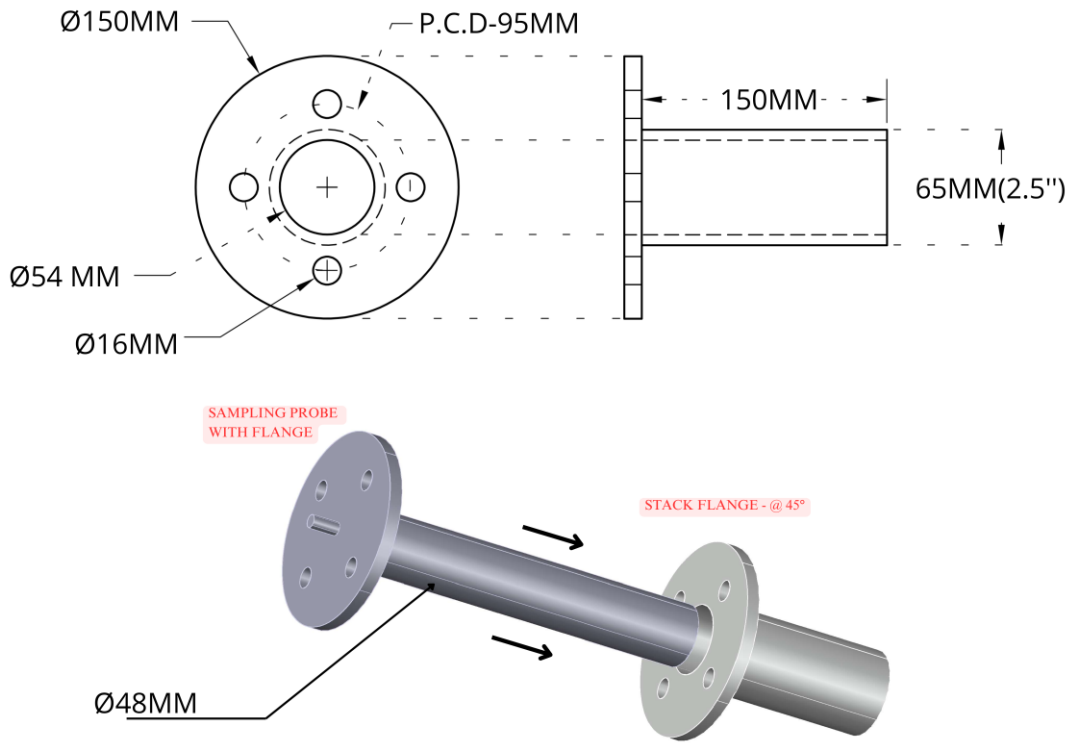


Fig. 3.1 Flange dimensions for CEMS sampling probe of Stack side

AADHAV INTECH
GST No: 33BORPA6650L1ZO
No: 60,3rd Street, Kasi Estate,
West Jafferkhanpet, Chennai – 600083.
Contact Number: 044 4204 1270
E Mail: service@aadhavintech.com

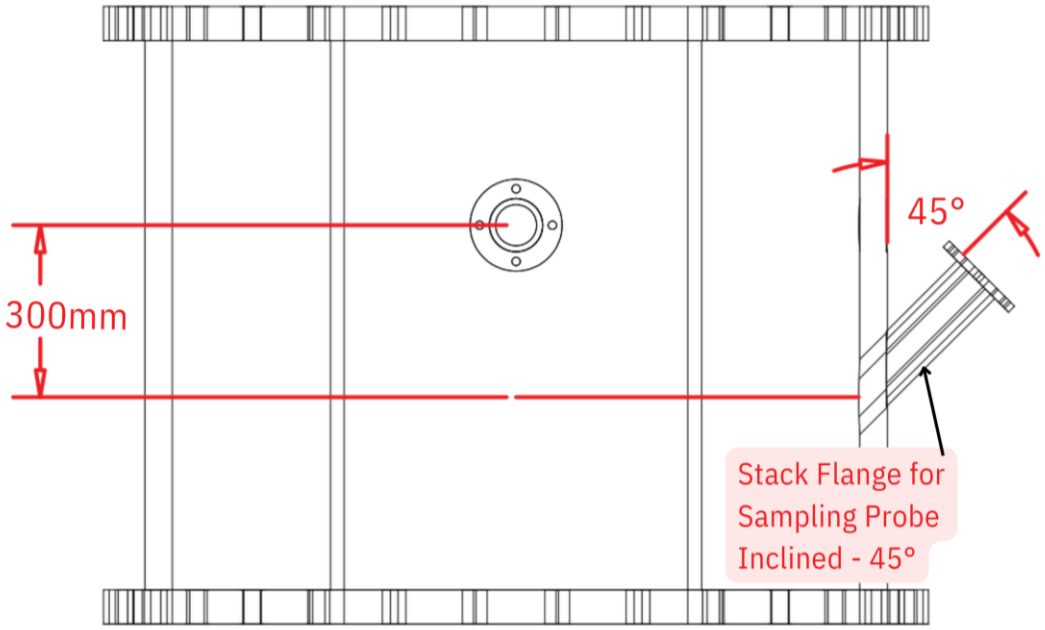


Fig. 3.2 CEMS Sampling Flange position 45° angle